NEET UG PHYSICS

Units and Measurement

Complete Exam-Oriented Tutorial: SI System, Significant Figures, Error Propagation, and Dimensional Analysis

1. Learning Objectives & Syllabus Overview

Physics is a quantitative science built upon the measurement of physical quantities. In the NEET UG examination, **Units and Measurement** forms the foundational bedrock for mechanics, electrodynamics, thermodynamics, and modern physics, while regularly contributing direct questions on significant figures, percentage errors, and dimensional analysis.

Key Learning Objectives:

- Understand fundamental (base) units, derived units, and the International System of Units (SI) including plane and solid angles.
- Convert physical quantities across unit systems (such as SI to CGS or custom hypothetical unit systems).
- Count significant figures accurately, apply rounding-off conventions, and perform arithmetic operations while preserving measurement precision.
- Calculate absolute, relative, and percentage uncertainties in arithmetic combinations of measured data.
- Determine dimensional formulae and dimensional equations of physical quantities.
- Apply the Principle of Homogeneity of Dimensions to check equation consistency and deduce mathematical relations among physical quantities.

2. Core Concepts and Explanations

2.1 Physical Quantities and Systems of Units

Measurement of any physical quantity involves comparing it with a basic, arbitrarily chosen, internationally accepted reference standard called a **unit**. Every measurement result is expressed by a numerical measure (n) accompanied by a unit (u).

- **Fundamental (or Base) Quantities & Units:** A limited set of independent physical quantities chosen as basic building blocks. Their units are called *base units*.
- **Derived Quantities & Units:** All other physical quantities whose units can be expressed as algebraic combinations of the base units.
- **System of Units:** A complete set of both base units and derived units.

Historically, three systems of units were used extensively for length, mass, and time:

- **CGS System:** centimetre (cm), gram (g), and second (s).
- **FPS (or British) System:** foot, pound, and second.
- **MKS System:** metre (m), kilogram (kg), and second (s).

2.2 The International System of Units (SI)

The internationally accepted system today is the *Système International d'Unités* (abbreviated as **SI**), developed by the International Bureau of Weights and Measures (BIPM) in 1971 and revised by the General Conference on Weights and Measures in November 2018. Because SI uses a decimal system, unit conversions within the system are simple and convenient.

Base Quantity	SI Unit Name	Symbol	Modern SI Definition (Fixed Fundamental Constants)
Length	metre	m	Defined by taking the fixed numerical value of the speed of light in vacuum c to be 299792458 when expressed in the unit $\mathrm{m\,s^{-1}}$, where the second is defined in terms of the caesium frequency $\Delta\nu_{\mathrm{Cs}}$.
Mass	kilogram	kg	Defined by taking the fixed numerical value of the Planck constant h to be $6.62607015 \times 10^{-34}$ when expressed in the unit $\mathrm{J\,s}$ (equal to $\mathrm{kg\,m^2\,s^{-1}}$), where the metre and second are defined in terms of c and $\Delta\nu_{\mathrm{Cs}}$.
Time	second	s	Defined by taking the fixed numerical value of the caesium frequency $\Delta\nu_{\mathrm{Cs}}$, the unperturbed ground-state hyperfine transition frequency of the caesium-133 atom, to be 9192631770 when expressed in the unit Hz (equal to $\mathrm{s^{-1}}$).
Electric current	ampere	A	Defined by taking the fixed numerical value of the elementary charge e to be $1.602176634 \times 10^{-19}$ when expressed in the unit C (equal to $\mathrm{A\,s}$), where the second is defined in terms of $\Delta\nu_{\mathrm{Cs}}$.
Thermodynamic temperature	kelvin	K	Defined by taking the fixed numerical value of the Boltzmann constant k to be 1.380649×10^{-23} when expressed in the unit $\mathrm{J\,K^{-1}}$ (equal to $\mathrm{kg\,m^2\,s^{-2}\,K^{-1}}$), where kg, m, and s are defined in terms of h, c, and $\Delta\nu_{\mathrm{Cs}}$.
Amount of substance	mole	mol	One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities (the fixed numerical value of the Avogadro constant N_{A} in $\mathrm{mol^{-1}}$). When the mole is used, the elementary entities (atoms, molecules, ions, electrons, or other specified particles) must be specified.
Luminous intensity	candela	cd	Defined by taking the fixed numerical value of the luminous efficacy of monochromatic radiation of frequency $540 \times 10^{12}\,\mathrm{Hz}$, K_{cd}, to be 683 when expressed in the unit $\mathrm{lm\,W^{-1}}$ (equal to $\mathrm{cd\,sr\,W^{-1}}$ or $\mathrm{cd\,sr\,kg^{-1}\,m^{-2}\,s^3}$).

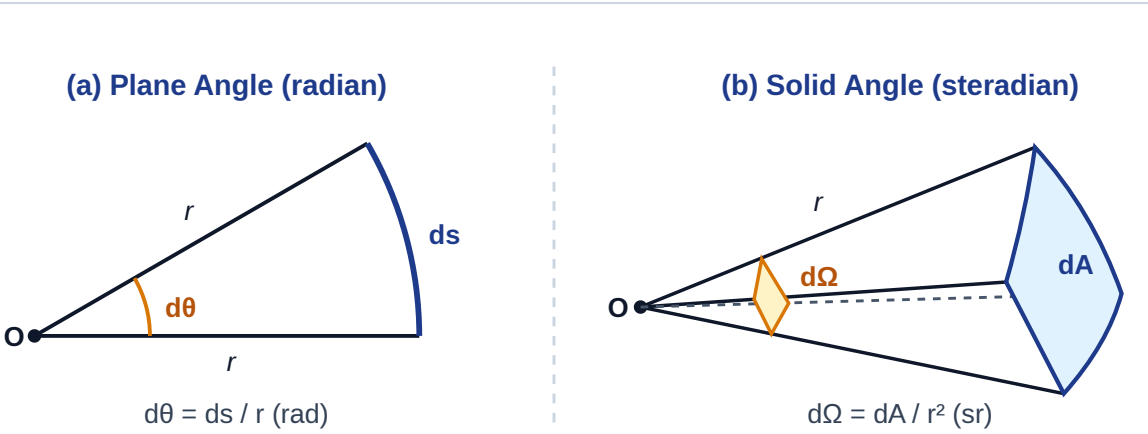
Note for NEET Aspirants: The exact multi-digit numerical values in the definitions above are provided to illustrate measurement precision and are not required to be memorized to nine digits; however, knowing which fundamental constant defines each SI base unit is an important conceptual takeaway.

Plane Angle and Solid Angle

In addition to the seven base units, two dimensionless quantities have special units in SI:

1. **Plane angle ($d\theta$):** The ratio of the length of arc ds to the radius r:
$$d\theta = \frac{ds}{r} \quad (\text{SI unit: radian, symbol: rad})$$
2. **Solid angle ($d\Omega$):** The ratio of the intercepted area dA of a spherical surface (described about the apex O as the centre) to the square of its radius r:
$$d\Omega = \frac{dA}{r^2} \quad (\text{SI unit: steradian, symbol: sr})$$

Both plane angle and solid angle are ratios of identical dimensions ($\mathrm{length/length}$ and $\mathrm{area/length^2}$), so **both are dimensionless quantities** even though they possess units (rad and sr).



Geometric representation of (a) plane angle $d\theta$ subtended by arc ds of radius r, and (b) solid angle $d\Omega$ subtended by spherical surface area element dA at radius r.

Units Retained for General Use (Outside SI)

Several non-SI units remain in common scientific and practical usage alongside SI units. Their exact values in SI units are summarized below:

Unit Name	Symbol	Value in SI Units	Physical Quantity
minute	min	$60\,\mathrm{s}$	Time
hour	h	$60\,\mathrm{min} = 3600\,\mathrm{s}$	Time
day	d	$24\,\mathrm{h} = 86400\,\mathrm{s}$	Time
year	y	$365.25\,\mathrm{d} = 3.156 \times 10^{7}\,\mathrm{s}$	Time
degree	$^\circ$	$1^\circ = \left(\frac{\pi}{180}\right)\mathrm{rad}$	Plane angle
litre	L	$1\,\mathrm{dm}^3 = 10^{-3}\,\mathrm{m}^3$	Volume
tonne	t	$10^{3}\,\mathrm{kg}$	Mass
quintal	q	$100\,\mathrm{kg}$	Mass
carat	c	$200\,\mathrm{mg} = 2 \times 10^{-4}\,\mathrm{kg}$	Mass
bar	bar	$0.1\,\mathrm{MPa} = 10^{5}\,\mathrm{Pa}$	Pressure
standard atmospheric pressure	atm	$101325\,\mathrm{Pa} = 1.013 \times 10^{5}\,\mathrm{Pa}$	Pressure
curie	Ci	$3.7 \times 10^{10}\,\mathrm{s}^{-1}$	Radioactivity
roentgen	R	$2.58 \times 10^{-4}\,\mathrm{C/kg}$	Exposure (ionizing radiation)
barn	b	$100\,\mathrm{fm}^2 = 10^{-28}\,\mathrm{m}^2$	Nuclear cross-sectional area
are	a	$1\,\mathrm{dam}^2 = 10^{2}\,\mathrm{m}^2$	Area
hectare	ha	$1\,\mathrm{hm}^2 = 10^{4}\,\mathrm{m}^2$	Area
ångström	$\mathrm{Å}$	$10^{-10}\,\mathrm{m}$	Atomic-scale length

2.3 Significant Figures

Every measurement involves errors, so the reported value must reflect the precision of the measuring instrument (which depends on its **least count**). The reported result of a measurement includes all digits that are known **reliably** plus the **first digit that is uncertain**. Together, the reliable digits plus the first uncertain digit are called **significant digits** or **significant figures**.

- If the period of oscillation of a simple pendulum is reported as $1.62\,\mathrm{s}$, the digits 1 and 6 are reliable and certain, while 2 is uncertain (3 significant figures).

- If the length of an object is reported as $287.5\,\mathrm{cm}$, the digits $2, 8, 7$ are certain while 5 is uncertain (4 significant figures).

Fundamental Principle of Unit Conversion: A choice of change of different units *does not* change the number of significant digits or figures in a measurement. For example:

$$2.308\,\mathrm{cm} = 0.02308\,\mathrm{m} = 23.08\,\mathrm{mm} = 23080\,\mu\mathrm{m}$$

All four representations have exactly **four** significant figures (the digits $2, 3, 0, 8$). The location of the decimal point is of no consequence in determining the number of significant figures.

Rules for Determining the Number of Significant Figures

1. **All non-zero digits are significant.** (e.g., 287.5 has 4 significant figures).
2. **All zeros between two non-zero digits are significant**, regardless of where the decimal point is located. (e.g., 2.005 and 2005 both have 4 significant figures).
3. **Leading zeros (zeros to the left of the first non-zero digit) are never significant.** In a number less than 1, zeros on the right of the decimal point but to the left of the first non-zero digit only serve to locate the decimal point. The zero conventionally placed to the left of a decimal point is also not significant. (e.g., in 0.002308, the three leading zeros are not significant; it has 4 significant figures).
4. **Trailing (terminal) zeros in a number without a decimal point are not significant.** (e.g., $123\,\mathrm{m} = 12300\,\mathrm{cm} = 123000\,\mathrm{mm}$ has 3 significant figures).
5. **Trailing zeros in a number with a decimal point are significant.** They explicitly convey the precision of measurement. (e.g., $3.500\,\mathrm{m}$ and $0.06900\,\mathrm{m}$ each have 4 significant figures).
6. **Exact numbers have an infinite number of significant figures.** Multiplying or dividing factors that are neither rounded numbers nor measured values—such as 2 in $r = d/2$ or $s = 2\pi r$, or the integer number of oscillations n in $T = t/n$—are exact and can be written as $2.0, 2.00, 2.0000$ as required.

Scientific Notation and Order of Magnitude

Converting a measured value like $4.700\,\mathrm{m}$ (4 significant figures) into millimetres as $4700\,\mathrm{mm}$ creates ambiguity because 4700 has no decimal point. To eliminate all ambiguity, measurements should be reported in **scientific notation**:

$$N = a \times 10^b$$

where a is the base number (mantissa) between 1 and 10, and b is any positive or negative integer exponent of 10.

- The power of 10 (10^b) is **irrelevant** to the determination of significant figures.
- All zeros appearing in the base number a in scientific notation (including trailing zeros) are **always significant**. Thus:
  $$4.700\,\mathrm{m} = 4.700 \times 10^2\,\mathrm{cm} = 4.700 \times 10^3\,\mathrm{mm} = 4.700 \times 10^{-3}\,\mathrm{km}$$
  unambiguously shows 4 significant figures in every unit.
- **Order of Magnitude:** To get an approximate idea of a number $a \times 10^b$, we round the base number a as follows:
  - If $a \leq 5$, round a to 1, so the number is approximately 10^b. The **order of magnitude** is b.
  - If $5 < a \leq 10$, round a to $10 = 10^1$, so the number is approximately 10^{b+1}. The **order of magnitude** is $b + 1$.

  Example: The diameter of the Earth ($1.28 \times 10^7\,\mathrm{m}$, where $1.28 \leq 5$) is of the order of $10^7\,\mathrm{m}$ (order of magnitude $= 7$). The diameter of a hydrogen atom ($1.06 \times 10^{-10}\,\mathrm{m}$, where $1.06 \leq 5$) is of the order of $10^{-10}\,\mathrm{m}$ (order of magnitude $= -10$). Thus, the diameter of the Earth is $7 - (-10) = 17$ orders of magnitude larger than the hydrogen atom.

2.4 Rules for Rounding Off Uncertain Digits

When dropping insignificant (uncertain) digits to round off a computation result to the required number of significant figures:

1. If the insignificant digit to be dropped is **greater than 5**, the preceding digit is **raised by 1**. (e.g., 2.746 rounded to 3 significant figures becomes 2.75).
2. If the insignificant digit to be dropped is **less than 5**, the preceding digit is **left unchanged**. (e.g., 1.743 rounded to 3 significant figures becomes 1.74).
3. If the insignificant digit to be dropped is **5**:
   - If the preceding digit is **even**, the insignificant digit is simply dropped (preceding digit remains unchanged). Example: 2.745 rounded to 3 significant figures becomes 2.74.
   - If the preceding digit is **odd**, the preceding digit is **raised by 1**. Example: 2.735 rounded to 3 significant figures becomes 2.74.

2.5 Rules for Arithmetic Operations with Significant Figures

A calculated result cannot be more accurate than the original measured values on which it is based.

- **Multiplication and Division Rule:** The final result should retain as many **significant figures** as there are in the original number with the *least significant figures*.

  Illustration: If mass $m = 4.237\ \text{g}$ (4 sig. figs.) and volume $V = 2.51\ \text{cm}^3$ (3 sig. figs.), then $\text{Density} = \dfrac{4.237\ \text{g}}{2.51\ \text{cm}^3} = 1.688047\ldots\ \text{g cm}^{-3}$, which must be rounded to 3 significant figures as $1.69\ \text{g cm}^{-3}$.

- **Addition and Subtraction Rule:** The final result should retain as many **decimal places** as there are in the number with the *least decimal places*.

  Illustration 1: $436.32\ \text{g} + 227.2\ \text{g} + 0.301\ \text{g} = 663.821\ \text{g}$. Since $227.2\ \text{g}$ has only 1 decimal place, the sum is rounded to 1 decimal place: $663.8\ \text{g}$.

  Illustration 2: $0.307\ \text{m} - 0.304\ \text{m} = 0.003\ \text{m} = 3 \times 10^{-3}\ \text{m}$ (both inputs have 3 decimal places, so the result is kept to 3 decimal places in metres, reducing the significant figures to 1).

Multi-step Computations: In complex multi-step calculations, always retain **one extra digit** beyond the significant digits in intermediate steps, and round off to the proper significant figures only at the end. For example, taking the reciprocal of 9.58 (3 sig. figs.): if rounded to 3 digits immediately as $1/9.58 = 0.104$, taking the reciprocal again gives $1/0.104 = 9.62$ (a rounding error). Retaining one extra digit in the intermediate step ($1/9.58 = 0.1044$) and then taking the reciprocal gives $1/0.1044 = 9.58$, recovering the exact original value.

2.6 Uncertainty in the Results of Arithmetic Calculations

1. **Product of Experimental Values (Combination of Percentage Errors):** Suppose the length and breadth of a thin rectangular sheet are measured with a metre scale (least count $0.1\ \text{cm}$) as:

   $$l = 16.2 \pm 0.1\ \text{cm} = 16.2\ \text{cm} \pm 0.6\%$$
   $$b = 10.1 \pm 0.1\ \text{cm} = 10.1\ \text{cm} \pm 1\%$$

   When multiplying l and b, the percentage errors add up:

   $$lb = 163.62\ \text{cm}^2 \pm (0.6\% + 1\%) = 163.62\ \text{cm}^2 \pm 1.6\% = 163.62 \pm 2.6\ \text{cm}^2$$

   Rounding the area to 3 significant figures (matching l and b) and the uncertainty accordingly gives:

   $$lb = 164 \pm 3\ \text{cm}^2$$

2. **Loss of Significant Figures in Subtraction:** Subtracting two nearly equal numbers reduces the number of significant figures. For example, $12.9\ \text{g} - 7.06\ \text{g} = 5.84\ \text{g}$ must be reported as $5.8\ \text{g}$ (1 decimal place, only 2 significant figures) even though both initial measurements had 3 significant figures.
3. **Dependence of Relative Error on the Magnitude of the Number:** Relative error depends not only on the number of significant figures n, but also on the number itself. Compare two mass measurements both

having 3 significant figures and least count ± 0.01 g:

- For $m_1 = 1.02 \pm 0.01$ g: $\text{Relative Error} = \left(\pm\frac{0.01}{1.02}\right) \times 100\% \approx \pm 1\%$.
- For $m_2 = 9.89 \pm 0.01$ g: $\text{Relative Error} = \left(\pm\frac{0.01}{9.89}\right) \times 100\% \approx \pm 0.1\%$.

2.7 Dimensions of Physical Quantities

The nature of a physical quantity is described by its **dimensions**. The seven base quantities are called the seven dimensions of the physical world and are denoted using square brackets $[\ ]$:

- Length: $[\mathrm{L}]$
- Mass: $[\mathrm{M}]$
- Time: $[\mathrm{T}]$
- Electric current: $[\mathrm{A}]$
- Thermodynamic temperature: $[\mathrm{K}]$
- Luminous intensity: $[\mathrm{cd}]$
- Amount of substance: $[\mathrm{mol}]$

Definition of Dimensions: The dimensions of a physical quantity are the *powers (or exponents)* to which the base quantities are raised to represent that quantity.

- **Dimensional Formula:** The expression showing how and which of the base quantities represent the dimensions of a physical quantity (e.g., $[\mathrm{M^0L^3T^0}]$ for volume, $[\mathrm{M^0LT^{-1}}]$ for speed or velocity, $[\mathrm{MLT^{-2}}]$ for force).
- **Dimensional Equation:** An equation obtained by equating a physical quantity with its dimensional formula:

$$[V] = [\mathrm{M^0L^3T^0}], \quad [v] = [\mathrm{M^0LT^{-1}}], \quad [F] = [\mathrm{MLT^{-2}}], \quad [\rho] = [\mathrm{ML^{-3}T^0}]$$

Note that dimensions capture the physical *quality* of a quantity, ignoring magnitude. Hence, initial velocity, final velocity, average velocity, change in velocity, and speed all have the same dimensional formula $[\mathrm{LT^{-1}}]$.

3. Essential Dimensional Formulae & Equations

Below is a high-yield reference table of dimensional formulae and SI units frequently tested in NEET UG:

Physical Quantity	Defining Relation	SI Unit	Dimensional Formula
Volume (V)	$\text{Length} \times \text{Breadth} \times \text{Height}$	$\mathrm{m^3}$	$[\mathrm{M^0L^3T^0}]$
Mass Density (ρ)	$\text{Mass}/\text{Volume}$	$\mathrm{kg\,m^{-3}}$	$[\mathrm{ML^{-3}T^0}]$
Speed / Velocity (v)	$\text{Length}/\text{Time}$	$\mathrm{m\,s^{-1}}$	$[\mathrm{M^0LT^{-1}}]$
Acceleration (a, g)	$\text{Velocity}/\text{Time}$	$\mathrm{m\,s^{-2}}$	$[\mathrm{M^0LT^{-2}}]$
Force (F)	$\text{Mass} \times \text{Acceleration}$	$\mathrm{N = kg\,m\,s^{-2}}$	$[\mathrm{MLT^{-2}}]$
Pressure / Stress (P)	Force/Area	$\mathrm{Pa = N\,m^{-2}}$	$[\mathrm{ML^{-1}T^{-2}}]$
Work / Energy / Heat (W, K, E)	$\text{Force} \times \text{Distance}$	$\mathrm{J = kg\,m^2\,s^{-2}}$	$[\mathrm{ML^2T^{-2}}]$
Power (P)	Work/Time	$\mathrm{W = J\,s^{-1}}$	$[\mathrm{ML^2T^{-3}}]$
Gravitational Constant (G)	$Fr^2/(m_1m_2)$	$\mathrm{N\,m^2\,kg^{-2}}$	$[\mathrm{M^{-1}L^3T^{-2}}]$
Planck Constant (h)	$\text{Energy}/\text{Frequency}$	$\mathrm{J\,s = kg\,m^2\,s^{-1}}$	$[\mathrm{ML^2T^{-1}}]$
Boltzmann Constant (k)	$\text{Energy}/\text{Temperature}$	$\mathrm{J\,K^{-1}}$	$[\mathrm{ML^2T^{-2}K^{-1}}]$
Electric Charge (q)	$\text{Current} \times \text{Time}$	$\mathrm{C = A\,s}$	$[\mathrm{M^0L^0TA}]$
Plane Angle (θ), Solid Angle (Ω), Refractive Index (μ), Relative Density	Ratio of like quantities	rad, sr, unitless	$[\mathrm{M^0L^0T^0}]$ (Dimensionless)

4. Methods and Applications of Dimensional Analysis

4.1 Application 1: Conversion of Units Between Two Systems

Since the magnitude of a physical quantity Q remains constant regardless of the system of units chosen, if n_1 and n_2 are the numerical values in two systems having units u_1 and u_2 respectively, and the dimensional formula of Q is $[\mathrm{M^aL^bT^c}]$, then:

$$Q = n_1 u_1 = n_2 u_2 \implies n_2 = n_1 \left[\frac{M_1}{M_2}\right]^a \left[\frac{L_1}{L_2}\right]^b \left[\frac{T_1}{T_2}\right]^c$$

4.2 Application 2: Checking the Dimensional Consistency of Equations

Principle of Homogeneity of Dimensions: The magnitudes of physical quantities may be added together or subtracted from one another *only* if they have the exact same dimensions. Consequently, in any valid physical equation $A = B + C - D$, the dimensions of every individual term on both sides must be identical:

$$[A] = [B] = [C] = [D]$$

Special Function Rule (High-Yield for NEET):

- The arguments of special mathematical functions—such as **trigonometric** ($\sin\theta, \cos\theta$), **logarithmic** ($\ln x$), and **exponential** (e^x) functions—must always be **dimensionless** $[\mathrm{M^0L^0T^0}]$.
- Pure numbers ($\pi, e, 1/2, 2$) and ratios of similar physical quantities (angle, refractive index, relative density) are dimensionless.

Illustration with Kinematic Equation: Consider $x = x_0 + v_0 t + \frac{1}{2}at^2$. Checking the dimensions of each term:

$$[x] = [\mathrm{L}], \quad [x_0] = [\mathrm{L}], \quad [v_0 t] = [\mathrm{LT^{-1}}][\mathrm{T}] = [\mathrm{L}], \quad \left[\frac{1}{2}at^2\right] = [\mathrm{LT^{-2}}][\mathrm{T^2}] = [\mathrm{L}]$$

Because every term on the right-hand side has the same dimension $[\mathrm{L}]$ as the left-hand side, the equation is dimensionally consistent.

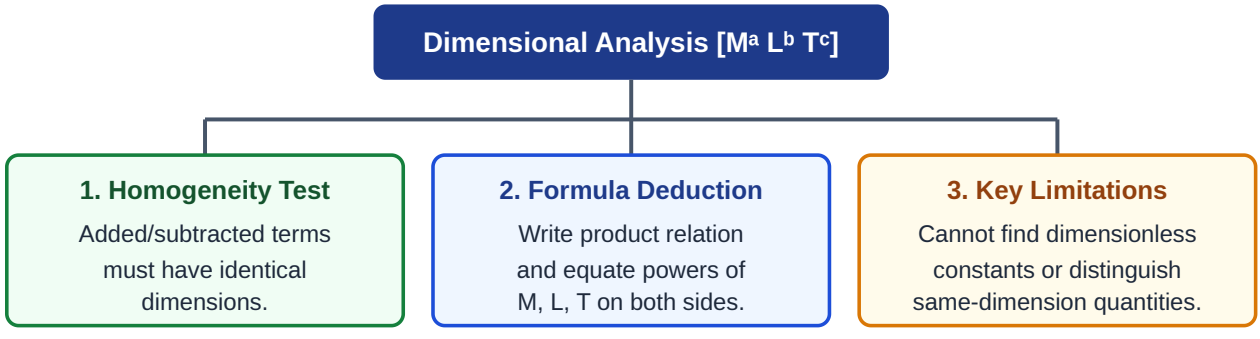
Crucial Caveat: A dimensionally consistent equation *need not* be an exact (actually correct) physical equation (due to unknown dimensionless constants like $1/2$), **but a dimensionally wrong or inconsistent equation must be wrong**.

4.3 Application 3: Deducing Relations Among Physical Quantities

If we know that a physical quantity Y depends on up to three linearly independent physical quantities P, Q, R via a product-type relationship, we can write:

$$Y = k\, P^x Q^y R^z$$

where k is a dimensionless constant. By substituting the dimensional formulae of Y, P, Q, R in terms of $[\mathrm{M}], [\mathrm{L}], [\mathrm{T}]$ and equating the exponents of each base dimension on both sides, we obtain a system of linear equations for x, y, z.



Summary of applications and inherent limitations of Dimensional Analysis.

4.4 Experimental Measurement Principles & Conceptual Connections

- **Comparing Precision of Length-Measuring Instruments:** The precision of an instrument is determined by its *least count* (the smallest value it can reliably measure):
  - *Vernier Callipers* with 20 divisions on the sliding vernier scale matching 19 main scale divisions (where $1\,\text{MSD} = 1\,\text{mm}$):
    $$\text{Least Count} = 1\,\text{MSD} - 1\,\text{VSD} = 1\,\text{mm} - \frac{19}{20}\,\text{mm} = \frac{1}{20}\,\text{mm} = 0.05\,\text{mm} = 5 \times 10^{-3}\,\text{cm}$$
  - *Screw Gauge* of pitch $1\,\text{mm}$ and 100 divisions on the circular scale:
    $$\text{Least Count} = \frac{\text{Pitch}}{\text{Number of circular scale divisions}} = \frac{1\,\text{mm}}{100} = 0.01\,\text{mm} = 10^{-3}\,\text{cm}$$
    Important Practical Limit: We cannot increase the accuracy of a screw gauge arbitrarily simply by increasing the number of divisions on the circular scale, because human eye resolution is limited and mechanical imperfections (such as non-uniformity of the screw threads and backlash) become dominant.
  - *Optical Instrument* measuring to within a wavelength of visible light:
    $$\text{Least Count} \approx \lambda \sim 4000\,\text{Å to } 7000\,\text{Å} \approx 5 \times 10^{-7}\,\text{m} = 5 \times 10^{-5}\,\text{cm}$$
    Thus, the optical instrument is the most precise among the three.

- **Measuring the Diameter of a Thin Thread with a Metre Scale:** Since a thin thread's diameter is smaller than the $1\,\text{mm}$ least count of a metre scale, wind the thread closely in n compact turns around a cylindrical pencil or rod so the turns touch each other without overlapping. Measure the total length l of the coiled turns using the metre scale; then the thread diameter is $d = \dfrac{l}{n}$.
- **Why a Larger Set of Measurements is More Reliable:** Random errors in experimental observations are equally likely to be positive or negative. Taking the arithmetic mean of 100 measurements reduces random error by a much larger factor than taking only 5 measurements, making the estimate of the true value much more reliable.
- **Relative Motion and Parallax (Train Window Observation):** When looking out of a fast-moving train, the line of sight to nearby objects (trees, houses) changes direction very rapidly (large angular shift per unit time), so they appear to move rapidly opposite to the train's motion. For very distant objects (hilltops, the Moon, stars), the distance is so large that the angle subtended by the train's displacement at the observer's eye is almost zero, making them appear stationary.

5. Comprehensive Worked Examples

Worked Example 1: Surface Area and Volume of a Cube with Significant Figures

Problem: Each side of a cube is measured to be $7.203\,\mathrm{m}$. What are the total surface area and the volume of the cube to appropriate significant figures?

Solution:

The measured side length $a = 7.203\,\mathrm{m}$ has **4 significant figures**. In multiplication, the calculated surface area and volume must also be rounded off to 4 significant figures (note that the factor 6 in the surface area formula is an exact integer with infinite significant figures).

1. **Total Surface Area:**

$$\text{Surface Area} = 6a^2 = 6 \times (7.203\,\mathrm{m})^2 = 311.299254\,\mathrm{m}^2$$

Rounding off to 4 significant figures gives:

$$\text{Surface Area} = 311.3\,\mathrm{m}^2$$

2. **Volume:**

$$\text{Volume} = a^3 = (7.203\,\mathrm{m})^3 = 373.714754\,\mathrm{m}^3$$

Rounding off to 4 significant figures gives:

$$\text{Volume} = 373.7\,\mathrm{m}^3$$

Worked Example 2: Density Calculation Using Significant Figures

Problem: $5.74\,\mathrm{g}$ of a substance occupies $1.2\,\mathrm{cm}^3$. Express its density by keeping the significant figures in view.

Solution:

The measured mass $m = 5.74\,\mathrm{g}$ has **3 significant figures**, whereas the measured volume $V = 1.2\,\mathrm{cm}^3$ has only **2 significant figures**. By the division rule, the density must be expressed to **2 significant figures**:

$$\text{Density} = \frac{5.74\,\mathrm{g}}{1.2\,\mathrm{cm}^3} = 4.7833\ldots\,\mathrm{g\,cm}^{-3} = 4.8\,\mathrm{g\,cm}^{-3}$$

Worked Example 3: Addition and Subtraction of Masses with Significant Figures

Problem: The mass of a box measured by a grocer's balance is $2.30\,\text{kg}$. Two gold pieces of masses $20.15\,\text{g}$ and $20.17\,\text{g}$ are added to the box. What is (a) the total mass of the box, and (b) the difference in the masses of the pieces to correct significant figures?

Solution:

(a) Total mass of the box: First express all masses in the same unit (kg):

- Mass of box $m_{\text{box}} = 2.30\,\text{kg}$ (2 decimal places in kg)
- Mass of first gold piece $m_1 = 20.15\,\text{g} = 0.02015\,\text{kg}$
- Mass of second gold piece $m_2 = 20.17\,\text{g} = 0.02017\,\text{kg}$

$$m_{\text{total}} = 2.30\,\text{kg} + 0.02015\,\text{kg} + 0.02017\,\text{kg} = 2.34032\,\text{kg}$$

In addition, the result must retain the same number of decimal places as the measurement with the least decimal places ($2.30\,\text{kg}$ has 2 decimal places). Rounding $2.34032\,\text{kg}$ to 2 decimal places gives:

$$m_{\text{total}} = 2.34\,\text{kg}$$

(b) Difference in the masses of the gold pieces:

$$\Delta m = 20.17\,\text{g} - 20.15\,\text{g} = 0.02\,\text{g}$$

Both measurements have 2 decimal places in grams, so the result is properly expressed to 2 decimal places as $0.02\,\text{g}$.

Worked Example 4: Area and Volume of a Rectangular Metal Sheet

Problem: The length, breadth, and thickness of a rectangular sheet of metal are $l = 4.234\,\text{m}$, $b = 1.005\,\text{m}$, and $t = 2.01\,\text{cm}$ respectively. Give the total surface area and volume of the sheet to correct significant figures.

Solution:

Convert all dimensions to metres:

- $l = 4.234\,\text{m}$ (4 significant figures)
- $b = 1.005\,\text{m}$ (4 significant figures)
- $t = 2.01\,\text{cm} = 0.0201\,\text{m}$ (3 significant figures)

Since the least precise measurement (t) has **3 significant figures**, both the final area and volume must be rounded to **3 significant figures**.

1. **Total Surface Area of the Rectangular Sheet:**

$$A = 2(lb + bt + tl) = 2\left[(4.234 \times 1.005) + (1.005 \times 0.0201) + (0.0201 \times 4.234)\right]\,\text{m}^2$$
$$A = 2\left[4.25517 + 0.0202005 + 0.0851034\right]\,\text{m}^2 = 2(4.3604739)\,\text{m}^2 = 8.7209478\,\text{m}^2$$

Rounding to 3 significant figures gives $A = 8.72\,\text{m}^2$.

2. **Volume of the Sheet:**

$$V = l \times b \times t = 4.234\,\text{m} \times 1.005\,\text{m} \times 0.0201\,\text{m} = 0.0855289\,\text{m}^3$$

Rounding to 3 significant figures gives $V = 0.0855\,\text{m}^3$.

Worked Example 5: Ruling Out Incorrect Formulae via Dimensional Consistency

Problem: The SI unit of energy is $\mathrm{J} = \mathrm{kg\,m^2\,s^{-2}}$; that of speed v is $\mathrm{m\,s^{-1}}$ and of acceleration a is $\mathrm{m\,s^{-2}}$. Which of the formulae for kinetic energy (K) given below can you rule out on the basis of dimensional arguments (m stands for the mass of the body)?

- (a) $K = m^2v^3$
- (b) $K = \frac{1}{2}mv^2$
- (c) $K = ma$
- (d) $K = \frac{3}{16}mv^2$
- (e) $K = \frac{1}{2}mv^2 + ma$

Solution:

From the SI unit of energy ($\mathrm{kg\,m^2\,s^{-2}}$), the dimensional formula of kinetic energy K is $[K] = [\mathrm{ML^2T^{-2}}]$. Let us evaluate the dimensions of the right-hand side (RHS) of each candidate equation:

- **(a)** $[m^2v^3] = [\mathrm{M}]^2[\mathrm{LT^{-1}}]^3 = [\mathrm{M^2L^3T^{-3}}] \neq [\mathrm{ML^2T^{-2}}]$. **Ruled out.**
- **(b)** $\left[\frac{1}{2}mv^2\right] = [\mathrm{M}][\mathrm{LT^{-1}}]^2 = [\mathrm{ML^2T^{-2}}]$. **Dimensionally consistent.**
- **(c)** $[ma] = [\mathrm{M}][\mathrm{LT^{-2}}] = [\mathrm{MLT^{-2}}] \neq [\mathrm{ML^2T^{-2}}]$. **Ruled out.**
- **(d)** $\left[\frac{3}{16}mv^2\right] = [\mathrm{M}][\mathrm{LT^{-1}}]^2 = [\mathrm{ML^2T^{-2}}]$. **Dimensionally consistent.**
- **(e)** In $\frac{1}{2}mv^2 + ma$, two quantities of different dimensions ($[\mathrm{ML^2T^{-2}}]$ and $[\mathrm{MLT^{-2}}]$) are added together, violating the Principle of Homogeneity. **Ruled out.**

Conclusion: Formulae **(a), (c), and (e)** are ruled out by dimensional arguments. Note that dimensional analysis alone cannot decide between (b) and (d) because both have valid dimensions; the actual factor $1/2$ comes from the physical definition of kinetic energy.

Worked Example 6: Deducing the Time Period of a Simple Pendulum

Problem: Consider a simple pendulum having a bob attached to a string that oscillates under the action of gravity. Suppose that the period of oscillation (T) depends on its length (l), mass of the bob (m), and acceleration due to gravity (g). Derive the expression for its time period using the method of dimensions.

Solution:

Write the product-type dependence of time period T on l, g, and m:

$$T = k\, l^{x} g^{y} m^{z}$$

where k is a dimensionless constant and x, y, z are exponents to be determined. Substituting the dimensions of each quantity on both sides:

$$[\mathrm{L}^{0}\mathrm{M}^{0}\mathrm{T}^{1}] = [\mathrm{L}^{1}]^{x}[\mathrm{L}^{1}\mathrm{T}^{-2}]^{y}[\mathrm{M}^{1}]^{z} = [\mathrm{L}^{x+y}\mathrm{M}^{z}\mathrm{T}^{-2y}]$$

Equating the powers of L, T, and M on both sides:

$$x + y = 0, \qquad -2y = 1, \qquad z = 0$$

Solving these linear equations gives:

$$y = -\frac{1}{2}, \qquad x = -y = \frac{1}{2}, \qquad z = 0$$

Substituting x, y, z back into the product relation:

$$T = k\, l^{1/2} g^{-1/2} m^{0} = k\sqrt{\frac{l}{g}}$$

(From mechanics, the dimensionless constant is $k = 2\pi$, giving $T = 2\pi\sqrt{l/g}$).

Worked Example 7: Conversion into a Custom System of Units (α, β, γ)

Problem: A calorie is a unit of heat (energy in transit) and it equals about $4.2\ \mathrm{J}$, where $1\ \mathrm{J} = 1\ \mathrm{kg\,m^{2}\,s^{-2}}$. Suppose we employ a system of units in which the unit of mass equals $\alpha\ \mathrm{kg}$, the unit of length equals $\beta\ \mathrm{m}$, and the unit of time is $\gamma\ \mathrm{s}$. Show that a calorie has a magnitude $4.2\,\alpha^{-1}\beta^{-2}\gamma^{2}$ in terms of the new units.

Solution:

The dimensional formula of energy is $[E] = [\mathrm{M}^{1}\mathrm{L}^{2}\mathrm{T}^{-2}]$. Let System 1 be SI units and System 2 be the new custom system:

- In SI (System 1): $n_1 = 4.2$, $M_1 = 1\ \mathrm{kg}$, $L_1 = 1\ \mathrm{m}$, $T_1 = 1\ \mathrm{s}$.
- In New System (System 2): $n_2 = ?$, $M_2 = \alpha\ \mathrm{kg}$, $L_2 = \beta\ \mathrm{m}$, $T_2 = \gamma\ \mathrm{s}$.

Using the unit conversion formula $n_2 = n_1\left[\frac{M_1}{M_2}\right]^{1}\left[\frac{L_1}{L_2}\right]^{2}\left[\frac{T_1}{T_2}\right]^{-2}$:

$$n_2 = 4.2\left[\frac{1\ \mathrm{kg}}{\alpha\ \mathrm{kg}}\right]^{1}\left[\frac{1\ \mathrm{m}}{\beta\ \mathrm{m}}\right]^{2}\left[\frac{1\ \mathrm{s}}{\gamma\ \mathrm{s}}\right]^{-2} = 4.2\,\alpha^{-1}\beta^{-2}\gamma^{2}$$

Thus, $1\ \mathrm{calorie} = 4.2\,\alpha^{-1}\beta^{-2}\gamma^{2}$ new units of energy.

Worked Example 8: Molar Volume vs. Atomic Volume & Solar Plasma Density

Problem: (a) The size (radius) of a hydrogen atom is about $0.5\ \text{Å}$ ($1\ \text{Å} = 10^{-10}\,\text{m}$). What is the total atomic volume in m^3 of a mole of hydrogen atoms? (b) One mole of an ideal gas at standard temperature and pressure (STP) occupies $22.4\,\text{L}$ (molar volume). Taking the size (diameter) of a hydrogen molecule to be about $1\ \text{Å}$ (so radius $r = 0.5\ \text{Å}$), find the ratio of molar volume to the atomic volume of a mole of hydrogen and explain why this ratio is so large. (c) Given the mass of the Sun $M = 2.0 \times 10^{30}\,\text{kg}$ and radius of the Sun $R = 7.0 \times 10^{8}\,\text{m}$, find the mass density of the Sun and state whether it lies in the range of solids/liquids or gases.

Solution:

(a) Atomic volume of 1 mole of hydrogen atoms: Treating each hydrogen atom as a sphere of radius $r = 0.5\ \text{Å} = 0.5 \times 10^{-10}\,\text{m}$, the volume of one atom is $V_1 = \frac{4}{3}\pi r^3$. For $1\,\text{mol}$ ($N_\text{A} = 6.023 \times 10^{23}$ atoms):

$$V_\text{atomic} = N_\text{A} \times \frac{4}{3}\pi r^3 = 6.023 \times 10^{23} \times \frac{4}{3} \times 3.14 \times (0.5 \times 10^{-10}\,\text{m})^3 = 3.15 \times 10^{-7}\,\text{m}^3$$

(b) Ratio of Molar Volume to Atomic Volume: Molar volume at STP is $V_\text{molar} = 22.4\,\text{L} = 22.4 \times 10^{-3}\,\text{m}^3$.

$$\frac{V_\text{molar}}{V_\text{atomic}} = \frac{22.4 \times 10^{-3}\,\text{m}^3}{3.15 \times 10^{-7}\,\text{m}^3} \approx 7.1 \times 10^4 \approx 10^4\ (\text{order of magnitude})$$

This ratio is very large because in a gas, the inter-atomic/inter-molecular separation is much larger than the size of an individual molecule, leaving vast empty space between molecules.

(c) Mass Density of the Sun:

$$\rho = \frac{M}{\frac{4}{3}\pi R^3} = \frac{2.0 \times 10^{30}\,\text{kg}}{\frac{4}{3} \times 3.14 \times (7.0 \times 10^{8}\,\text{m})^3} \approx 1.39 \times 10^3\,\text{kg}\,\text{m}^{-3} = 1.4 \times 10^3\,\text{kg}\,\text{m}^{-3}$$

Even though the Sun is a hot ionized plasma at temperatures of $10^7\,\text{K}$ (core) to $6000\,\text{K}$ (surface), the intense inward gravitational attraction of its huge mass compresses the plasma so strongly that its density ($1.4 \times 10^3\,\text{kg}\,\text{m}^{-3}$) is in the range of densities of **solids and liquids**, not gases.

6. NEET UG Practice Questions with Complete Solutions

The following original practice questions are strictly aligned with the NEET UG syllabus and pattern to test your mastery of the concepts taught in this tutorial.

Practice Question 1 (Unit Conversions & Derived Units)

Match List-I (Physical Quantities with Given Values) with List-II (Equivalent Values in Target Units) and select the correct option:

List-I	List-II
(A) Volume of a cube of side $1\,\text{cm}$	(I) $3.9 \times 10^4\,\text{km}\,\text{h}^{-2}$
(B) Total surface area of a solid cylinder of radius $2.0\,\text{cm}$ and height $10.0\,\text{cm}$	(II) $6.67 \times 10^{-8}\,\text{cm}^3\,\text{s}^{-2}\,\text{g}^{-1}$
(C) Acceleration of $3.0\,\text{m}\,\text{s}^{-2}$	(III) $1.5 \times 10^4\,\text{mm}^2$
(D) Universal gravitational constant $G = 6.67 \times 10^{-11}\,\text{N}\,\text{m}^2\,\text{kg}^{-2}$	(IV) $10^{-6}\,\text{m}^3$

(1) (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

(2) (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

(3) (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

(4) (A)-(IV), (B)-(III), (C)-(II), (D)-(I)

Correct Answer: (1)

Worked Solution:

- **(A)** Side $a = 1\,\text{cm} = 10^{-2}\,\text{m}$. Volume $V = (10^{-2}\,\text{m})^3 = 10^{-6}\,\text{m}^3 \rightarrow$ **(IV)**.
- **(B)** For a solid cylinder, $r = 2.0\,\text{cm} = 20\,\text{mm}$ and $h = 10.0\,\text{cm} = 100\,\text{mm}$. Total surface area $S = 2\pi r(r + h) = 2 \times 3.14 \times 20 \times (20 + 100)\,\text{mm}^2 = 15072\,\text{mm}^2$. To 2 significant figures, $S = 1.5 \times 10^4\,\text{mm}^2 \rightarrow$ **(III)**.
- **(C)** $1\,\text{m} = 10^{-3}\,\text{km}$ and $1\,\text{s} = \dfrac{1}{3600}\,\text{h}$. Thus: \[3.0\,\mathrm{m\,s^{-2}} = 3.0 \times \frac{10^{-3}\,\mathrm{km}}{(1/3600\,\mathrm{h})^2} = 3.0 \times 10^{-3} \times (3.6 \times 10^3)^2\,\mathrm{km\,h^{-2}} = 3.888 \times 10^4\,\mathrm{km\,h^{-2}} \approx 3.9 \times 10^4\,\mathrm{km\,h^{-2}} \rightarrow \text{**(I)**}. \]
- **(D)** Since $1\,\text{N} = 1\,\text{kg}\,\text{m}\,\text{s}^{-2}$, we have $1\,\text{N}\,\text{m}^2\,\text{kg}^{-2} = 1\,\text{m}^3\,\text{s}^{-2}\,\text{kg}^{-1} = \dfrac{(10^2\,\text{cm})^3}{\text{s}^2\,(10^3\,\text{g})} = 10^3\,\text{cm}^3\,\text{s}^{-2}\,\text{g}^{-1}$. Therefore: \[G = 6.67 \times 10^{-11} \times 10^3\,\mathrm{cm^3\,s^{-2}\,g^{-1}} = 6.67 \times 10^{-8}\,\mathrm{cm^3\,s^{-2}\,g^{-1}} \rightarrow \text{**(II)**}. \]

Practice Question 2 (Rules for Counting Significant Figures)

Consider the following six physical measurements:

(i) $0.007\,\mathrm{m^2}$ (ii) $2.64 \times 10^{24}\,\mathrm{kg}$ (iii) $0.2370\,\mathrm{g\,cm^{-3}}$ (iv) $6.320\,\mathrm{J}$ (v) $6.032\,\mathrm{N\,m^{-2}}$ (vi) $0.0006032\,\mathrm{m^2}$

The respective number of significant figures in (i), (ii), (iii), (iv), (v), and (vi) are:

(1) 3, 3, 4, 4, 4, 7

(2) 1, 3, 4, 4, 4, 4

(3) 1, 3, 3, 3, 4, 4

(4) 3, 24, 4, 4, 4, 8

Correct Answer: (2)

Worked Solution:

- (i) $0.007\,\mathrm{m^2}$: Number is less than 1; zeros to the right of the decimal point but to the left of the first non-zero digit (7) are not significant $\rightarrow$ **1** significant figure.
- (ii) $2.64 \times 10^{24}\,\mathrm{kg}$: The power of 10 (10^{24}) is irrelevant; all digits in 2.64 are significant $\rightarrow$ **3** significant figures.
- (iii) $0.2370\,\mathrm{g\,cm^{-3}}$: Trailing zero in a number with a decimal point is significant $(2, 3, 7, 0) \rightarrow$ **4** significant figures.
- (iv) $6.320\,\mathrm{J}$: Trailing zero after the decimal point is significant $(6, 3, 2, 0) \rightarrow$ **4** significant figures.
- (v) $6.032\,\mathrm{N\,m^{-2}}$: Zero trapped between non-zero digits is significant $(6, 0, 3, 2) \rightarrow$ **4** significant figures.
- (vi) $0.0006032\,\mathrm{m^2}$: Leading zeros are not significant; the digits $6, 0, 3, 2$ are significant $\rightarrow$ **4** significant figures.

Practice Question 3 (Dimensional Consistency in Relativistic Formula)

A famous relation in physics relates the 'moving mass' m to the 'rest mass' m_0 of a particle in terms of its speed v and the speed of light c. A student recalls the relation almost correctly but forgets where to put the constant c, writing:

$$m = \frac{m_0}{(1 - v^2)^{1/2}}$$

Using the Principle of Homogeneity of Dimensions, the correct expression after inserting c is:

(1) $m = \dfrac{m_0 c}{(1 - v^2)^{1/2}}$

(2) $m = \dfrac{m_0}{(1 - v^2 c^2)^{1/2}}$

(3) $m = \dfrac{m_0}{\left(1 - \dfrac{v^2}{c^2}\right)^{1/2}}$

(4) $m = \dfrac{m_0}{\left(1 - \dfrac{c^2}{v^2}\right)^{1/2}}$

Correct Answer: (3)

Worked Solution:

On the left-hand side, $[m] = [\mathrm{M}]$, and in the numerator on the right-hand side, $[m_0] = [\mathrm{M}]$. Therefore, the denominator $(1 - \ldots)^{1/2}$ must be completely **dimensionless** $[\mathrm{M}^0\mathrm{L}^0\mathrm{T}^0]$. Since 1 is a dimensionless pure number, by the Principle of Homogeneity of Dimensions, the term subtracted from 1 must also be dimensionless. Because $[v^2] = [\mathrm{L}^2\mathrm{T}^{-2}]$ and $[c^2] = [\mathrm{L}^2\mathrm{T}^{-2}]$, v^2 must be divided by c^2 to make $\dfrac{v^2}{c^2}$ dimensionless (and less than 1 for $v < c$). Thus, the correct formula is $m = \dfrac{m_0}{(1 - v^2/c^2)^{1/2}}$.

Practice Question 4 (Custom Unit of Length Based on Speed of Light)

A new unit of length is chosen such that the speed of light in vacuum is unity ($1\,\text{new unit of length} \cdot \text{s}^{-1}$). What is the distance between the Sun and the Earth in terms of this new unit if light takes $8\,\text{min}$ and $20\,\text{s}$ to cover this distance?

(1) 8.33 new units

(2) 500 new units

(3) 1.5×10^{11} new units

(4) 250 new units

Correct Answer: (2)

Worked Solution:

Given speed of light in the new system: $c = 1\,\text{new unit of length} \cdot \text{s}^{-1}$.

Time taken by light to travel from the Sun to the Earth:

$$t = 8\,\text{min} + 20\,\text{s} = (8 \times 60)\,\text{s} + 20\,\text{s} = 480\,\text{s} + 20\,\text{s} = 500\,\text{s}$$

Therefore, the distance between the Sun and the Earth is:

$$d = c \times t = \left(1\,\text{new unit of length} \cdot \text{s}^{-1}\right) \times 500\,\text{s} = 500\,\text{new units of length}$$

Practice Question 5 (Magnification in Optical Measurements)

The photograph of a house occupies an area of $1.75\ \text{cm}^2$ on a $35\ \text{mm}$ slide. The slide is projected onto a screen, and the area of the house on the screen is $1.55\ \text{m}^2$. What is the linear magnification of the projector-screen arrangement (to appropriate significant figures)?

(1) 88.6

(2) 94.1

(3) 8857

(4) 9.41

Correct Answer: (2)

Worked Solution:

Area of the object (on the slide): $A_o = 1.75\ \text{cm}^2$.

Area of the image (on the screen): $A_i = 1.55\ \text{m}^2 = 1.55 \times (10^2\ \text{cm})^2 = 1.55 \times 10^4\ \text{cm}^2$.

Areal magnification is given by:

$$m_{\text{areal}} = \frac{A_i}{A_o} = \frac{1.55 \times 10^4\ \text{cm}^2}{1.75\ \text{cm}^2} = 8857.14$$

Since linear magnification $m_{\text{linear}} = \sqrt{m_{\text{areal}}}$:

$$m_{\text{linear}} = \sqrt{8857.14} \approx 94.11 \approx 94.1 \quad \text{(rounded to 3 significant figures)}$$

Practice Question 6 (Microscope Magnification & Thickness Estimation)

A student measures the thickness of a human hair by looking at it through a microscope of magnification 100. He makes 20 observations and finds that the average width of the hair in the field of view of the microscope is $3.5\ \text{mm}$. What is the estimate on the actual thickness of the hair?

(1) $0.035\ \text{mm}$

(2) $0.35\ \text{mm}$

(3) $1.75\ \mu\text{m}$

(4) $350\ \text{mm}$

Correct Answer: (1)

Worked Solution:

Linear magnification of the microscope is defined as:

$$\text{Magnification } (M) = \frac{\text{Apparent (observed) width}}{\text{Actual (real) thickness}}$$

Given $M = 100$ and average observed width $= 3.5\ \text{mm}$ (the number of observations 20 was used to compute this reliable average):

$$\text{Actual thickness} = \frac{3.5\ \text{mm}}{100} = 0.035\ \text{mm} = 35\ \mu\text{m}$$

Practice Question 7 (Rounding Off & Arithmetic Rules)

Two numbers $A = 2.745$ and $B = 2.735$ are each rounded off to three significant figures using standard scientific rounding conventions. What are the rounded values of A and B respectively?

(1) 2.75 and 2.74

(2) 2.74 and 2.73

(3) 2.74 and 2.74

(4) 2.75 and 2.73

Correct Answer: (3)

Worked Solution:

When the insignificant digit to be dropped is 5:

- In $A = 2.745$, the digit preceding 5 is 4 (which is **even**), so 5 is simply dropped and 4 remains unchanged $\rightarrow 2.74$.
- In $B = 2.735$, the digit preceding 5 is 3 (which is **odd**), so the preceding digit is raised by $1 \rightarrow 2.74$.

Hence, both A and B round off to 2.74.

Practice Question 8 (Assertion–Reason on Dimensional Analysis)

Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.

Assertion (A): Both plane angle (rad) and solid angle (sr) have SI units, yet both are dimensionless physical quantities.

Reason (R): Plane angle is defined as the ratio of arc length to radius ($d\theta = ds/r$), and solid angle is defined as the ratio of intercepted spherical area to the square of radius ($d\Omega = dA/r^2$), so the dimensions of length cancel out in both ratios.

In the light of the above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

Correct Answer: (1)

Worked Solution:

Since $[d\theta] = \dfrac{[ds]}{[r]} = \dfrac{[\mathrm{L}]}{[\mathrm{L}]} = [\mathrm{M}^0\mathrm{L}^0\mathrm{T}^0]$ and $[d\Omega] = \dfrac{[dA]}{[r^2]} = \dfrac{[\mathrm{L}^2]}{[\mathrm{L}^2]} = [\mathrm{M}^0\mathrm{L}^0\mathrm{T}^0]$, both quantities are dimensionless ratios of like lengths/areas, even though they have the supplementary SI units radian (rad) and steradian (sr). Thus both (A) and (R) are true and (R) correctly explains (A).

Practice Question 9 (Order of Magnitude Comparison)

The diameter of the Earth is $1.28 \times 10^7\,\mathrm{m}$ and the diameter of a hydrogen atom is $1.06 \times 10^{-10}\,\mathrm{m}$. If the speed of light in vacuum is $3.00 \times 10^8\,\mathrm{m\,s^{-1}}$ and $1\,\mathrm{year} = 365.25\,\mathrm{d} = 3.1557 \times 10^7\,\mathrm{s}$, which of the following statements is completely correct?

(1) The diameter of the Earth is 17 orders of magnitude larger than the hydrogen atom, and $1\,\mathrm{light\ year} = 9.47 \times 10^{15}\,\mathrm{m}$ (3 significant figures).

(2) The diameter of the Earth is 16 orders of magnitude larger than the hydrogen atom, and $1\,\mathrm{light\ year} = 9.4671 \times 10^{15}\,\mathrm{m}$ (5 significant figures).

(3) $1\,\mathrm{m} \approx 9.47 \times 10^{-16}\,\mathrm{ly}$ and the order of magnitude of the Earth's diameter is 8.

(4) Both (1) and (3) are correct.

Correct Answer: (1)

Worked Solution:

- For Earth's diameter $1.28 \times 10^7\,\mathrm{m}$, since $1.28 \leq 5$, the order of magnitude is 7.
- For the hydrogen atom's diameter $1.06 \times 10^{-10}\,\mathrm{m}$, since $1.06 \leq 5$, the order of magnitude is -10.
- Difference in orders of magnitude $= 7 - (-10) = 17$.
- Distance of $1\,\mathrm{light\ year} = c \times t = (3.00 \times 10^8\,\mathrm{m\,s^{-1}}) \times (3.1557 \times 10^7\,\mathrm{s}) = 9.4671 \times 10^{15}\,\mathrm{m}$. Since 3.00×10^8 has 3 significant figures and 3.1557×10^7 has 5 significant figures, the product must be rounded to 3 significant figures: $9.47 \times 10^{15}\,\mathrm{m}$ (and $1\,\mathrm{m} = \dfrac{1}{9.47 \times 10^{15}}\,\mathrm{ly} = 1.06 \times 10^{-16}\,\mathrm{ly}$).

Practice Question 10 (Relative Density & Unit Conversion)

A vehicle moving with a speed of $18\,\mathrm{km\,h^{-1}}$ covers $x\,\mathrm{m}$ in $1\,\mathrm{s}$. The relative density of lead is 11.3, so its density in SI units is $y\,\mathrm{kg\,m^{-3}}$. The values of x and y are respectively:

(1) $x = 5, y = 11.3$

(2) $x = 5, y = 1.13 \times 10^4$

(3) $x = 64.8, y = 1.13 \times 10^4$

(4) $x = 50, y = 11.3 \times 10^2$

Correct Answer: (2)

Worked Solution:

- Speed of the vehicle: $v = 18\,\mathrm{km\,h^{-1}} = 18 \times \dfrac{1000\,\mathrm{m}}{3600\,\mathrm{s}} = 5\,\mathrm{m\,s^{-1}}$. Thus in $1\,\mathrm{s}$, it covers $x = 5\,\mathrm{m}$.
- Relative density of lead $= \dfrac{\text{Density of lead}}{\text{Density of water at } 4\,^\circ\mathrm{C}} = 11.3$. Since the density of water is $1\,\mathrm{g\,cm^{-3}} = 10^3\,\mathrm{kg\,m^{-3}}$, the density of lead is $11.3\,\mathrm{g\,cm^{-3}} = 11.3 \times 10^3\,\mathrm{kg\,m^{-3}} = 1.13 \times 10^4\,\mathrm{kg\,m^{-3}}$.

7. Common Pitfalls & NEET Revision Checklist

Common Exam Pitfalls to Avoid:

1. **Confusing Addition/Subtraction Rules with Multiplication/Division Rules:** In addition and subtraction, look ONLY at the number of *decimal places* (not total significant figures). In multiplication and division, look ONLY at the total number of *significant figures*.
2. **Counting Powers of 10 or Leading Zeros as Significant:** Neither the 10^b factor in scientific notation nor leading zeros in decimals like 0.006032 are ever significant.
3. **Assuming Dimensional Correctness Guarantees Physical Correctness:** An equation like $s = ut + 5at^2$ is dimensionally consistent, yet physically wrong because the dimensionless constant is $1/2$, not 5. However, a dimensionally inconsistent equation is *always* wrong.
4. **Forgetting that Angles Have Units Vision but No Dimensions:** Plane angle (rad) and solid angle (sr) are dimensionless ($[\mathrm{M^0L^0T^0}]$), whereas constants like Planck's constant h ($[\mathrm{ML^2T^{-1}}]$) and Gravitational constant G ($[\mathrm{M^{-1}L^3T^{-2}}]$) are dimensional constants.
5. **Calling a Dimensional Quantity 'Large' or 'Small' Without a Standard:** Statements like "atoms are very small" or "a jet plane moves with great speed" are meaningless unless compared to a reference standard of the same dimension (e.g., "atoms are much smaller than the tip of a sharp pin", whereas "a proton is much more massive than an electron" is already a valid comparison between two masses).

8. Concise Final Recap

- **SI System:** Built on 7 base units $(\mathrm{m, kg, s, A, K, mol, cd})$ plus 2 dimensionless supplementary angle units $(\mathrm{rad, sr})$.

- **Significant Figures:** Reliable digits $+$ first uncertain digit. Unchanged by unit conversions; best expressed in scientific notation $a \times 10^b$.
- **Arithmetic Precision:** Multiplication/Division $\rightarrow$ least significant figures; Addition/Subtraction $\rightarrow$ least decimal places. Retain 1 extra digit in intermediate steps of multi-step calculations.
- **Dimensional Analysis:** Tests dimensional homogeneity ($[A] = [B]$ for $A \pm B$), converts units via $n_2 = n_1(M_1/M_2)^a(L_1/L_2)^b(T_1/T_2)^c$, and deduces product-form physical relationships.